

Run MATRIX procedure:

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This version of PROCESS requires SPSS version 26 or later

Workshop schedule available at [haskayne.ucalgary.ca/CCRAM](http://haskayne.ucalgary.ca/CCRAM)

In SPSS 29 and later, change default output font to Courier New for tidier output. More information about PROCESS at [processmacro.org/faq.html](http://processmacro.org/faq.html).

This beta release has not been completely tested. Use at your own risk.

\*\*\*\*\* PROCESS Procedure for SPSS Version 5.0 \*\*\*\*\*

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Documentation available in Hayes (2022). [www.guilford.com/p/hayes3](http://www.guilford.com/p/hayes3)

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Model: 4  
Y: engage  
X: psycap  
M: jobsat

Sample  
Size: 270

#### Variable descriptive statistics

|      | engage | psycap | jobsat |
|------|--------|--------|--------|
| Mean | 4.2263 | 4.3211 | 3.6941 |
| SD   | .9087  | .8245  | .6913  |
| Min  | 1.4444 | 1.5833 | 1.6500 |
| Max  | 6.0000 | 5.9583 | 4.9500 |

#### Variable intercorrelations (Pearson r)

|        | engage | psycap | jobsat |
|--------|--------|--------|--------|
| engage | 1.0000 | .5326  | .6509  |
| psycap | .5326  | 1.0000 | .4850  |
| jobsat | .6509  | .4850  | 1.0000 |

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OUTCOME VARIABLE:  
jobsat

#### Model Summary

| R     | R-sq  | MSE   | F       | df1    | df2      | p     |
|-------|-------|-------|---------|--------|----------|-------|
| .4850 | .2352 | .3669 | 82.4165 | 1.0000 | 268.0000 | .0000 |

#### Model

| coeff | se | t | p | LLCI | ULCI |
|-------|----|---|---|------|------|
|-------|----|---|---|------|------|

|          |        |       |        |       |        |        |
|----------|--------|-------|--------|-------|--------|--------|
| constant | 1.9368 | .1970 | 9.8297 | .0000 | 1.5489 | 2.3248 |
| psycap   | .4067  | .0448 | 9.0784 | .0000 | .3185  | .4949  |

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Some regression diagnostics

|          |         |        |
|----------|---------|--------|
|          | Min.    | Max.   |
| fitted   | 2.5807  | 4.3599 |
| residual | -1.9210 | 1.3729 |
| t-resid  | -3.2388 | 2.2926 |

Shape of residuals

|       |          |          |
|-------|----------|----------|
|       | Skewness | Kurtosis |
| Value | -.3691   | -.0489   |
| se    | .1483    | .2954    |

Bonferroni-corrected p for largest t-residual

|         |         |          |
|---------|---------|----------|
| t-resid | p-value | casenum  |
| -3.2388 | .3651   | 113.0000 |

Most influential observations

|          |          |        |
|----------|----------|--------|
|          | casenum  | dfbeta |
| constant | 139.0000 | -.0666 |
| psycap   | 139.0000 | .0146  |

Variable tolerance and VIF

|        |        |        |
|--------|--------|--------|
|        | Tol.   | VIF    |
| psycap | 1.0000 | 1.0000 |

Breusch-Pagan test of heteroskedasticity

|        |        |        |       |
|--------|--------|--------|-------|
|        | Chi-sq | df     | p     |
| Normal | .0091  | 1.0000 | .9241 |
| Robust | .0094  | 1.0000 | .9227 |

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OUTCOME VARIABLE:

engage

Model Summary

|       |       |       |          |        |          |       |
|-------|-------|-------|----------|--------|----------|-------|
| R     | R-sq  | MSE   | F        | df1    | df2      | p     |
| .6966 | .4852 | .4282 | 125.8437 | 2.0000 | 267.0000 | .0000 |

Model

|          |       |       |         |       |        |       |
|----------|-------|-------|---------|-------|--------|-------|
|          | coeff | se    | t       | p     | LLCI   | ULCI  |
| constant | .3828 | .2483 | 1.5417  | .1243 | -.1061 | .8717 |
| psycap   | .3126 | .0553 | 5.6486  | .0000 | .2036  | .4215 |
| jobsat   | .6748 | .0660 | 10.2255 | .0000 | .5449  | .8048 |

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Some regression diagnostics

|          | Min.    | Max.   |
|----------|---------|--------|
| fitted   | 1.9912  | 5.5181 |
| residual | -1.4015 | 2.1740 |
| t-resid  | -2.1850 | 3.4242 |

Shape of residuals

|       | Skewness | Kurtosis |
|-------|----------|----------|
| Value | .0002    | -.1889   |
| se    | .1483    | .2954    |

Bonferroni-corrected p for largest t-residual

| t-resid | p-value | casenum  |
|---------|---------|----------|
| 3.4242  | .1928   | 261.0000 |

Most influential observations

|          | casenum  | dfbeta |
|----------|----------|--------|
| constant | 261.0000 | .1157  |
| psycap   | 74.0000  | .0170  |
| jobsat   | 113.0000 | -.0236 |

Variable tolerance and VIF

|        | Tol.  | VIF    |
|--------|-------|--------|
| psycap | .7648 | 1.3075 |
| jobsat | .7648 | 1.3075 |

Breusch-Pagan test of heteroskedasticity

|        | Chi-sq | df     | p     |
|--------|--------|--------|-------|
| Normal | 8.3005 | 2.0000 | .0158 |
| Robust | 9.2617 | 2.0000 | .0097 |

\*\*\*\*\* TOTAL EFFECT MODEL \*\*\*\*\*

OUTCOME VARIABLE:

engage

Model Summary

| R     | R-sq  | MSE   | F        | df1    | df2      | p     |
|-------|-------|-------|----------|--------|----------|-------|
| .5326 | .2837 | .5937 | 106.1195 | 1.0000 | 268.0000 | .0000 |

Model

|          | coeff  | se    | t       | p     | LLCI   | ULCI   |
|----------|--------|-------|---------|-------|--------|--------|
| constant | 1.6898 | .2507 | 6.7417  | .0000 | 1.1963 | 2.1833 |
| psycap   | .5870  | .0570 | 10.3014 | .0000 | .4748  | .6992  |

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Some regression diagnostics

|          | Min.    | Max.   |
|----------|---------|--------|
| fitted   | 2.6192  | 5.1874 |
| residual | -2.4515 | 1.8958 |
| t-resid  | -3.2490 | 2.4981 |

#### Shape of residuals

|       | Skewness | Kurtosis |
|-------|----------|----------|
| Value | -.2517   | -.1072   |
| se    | .1483    | .2954    |

#### Bonferroni-corrected p for largest t-residual

| t-resid | p-value | Casenum |
|---------|---------|---------|
| -3.2490 | .3527   | 24.0000 |

#### Most influential observations

|          | casenum  | dfbeta |
|----------|----------|--------|
| constant | 241.0000 | -.0835 |
| psycap   | 241.0000 | .0179  |

#### Variable tolerance and VIF

|        | Tol.   | VIF    |
|--------|--------|--------|
| psycap | 1.0000 | 1.0000 |

#### Breusch-Pagan test of heteroskedasticity

|        | Chi-sq | df     | p     |
|--------|--------|--------|-------|
| Normal | 1.3055 | 1.0000 | .2532 |
| Robust | 1.3943 | 1.0000 | .2377 |

#### \*\*\*\*\* TOTAL, DIRECT, AND INDIRECT EFFECTS OF X ON Y \*\*\*\*\*

##### Total effect of X on Y

| Effect | se    | t       | p     | LLCI  | ULCI  |
|--------|-------|---------|-------|-------|-------|
| .5870  | .0570 | 10.3014 | .0000 | .4748 | .6992 |

##### Direct effect of X on Y

| Effect | se    | t      | p     | LLCI  | ULCI  |
|--------|-------|--------|-------|-------|-------|
| .3126  | .0553 | 5.6486 | .0000 | .2036 | .4215 |

##### Indirect effect(s) of X on Y:

|        | Effect | BootSE | BootLLCI | BootULCI |
|--------|--------|--------|----------|----------|
| jobsat | .2744  | .0413  | .1992    | .3597    |

##### Cases with greatest influence on indirect effect(s):

|        | casenum  | dfb_ie |
|--------|----------|--------|
| jobsat | 113.0000 | -.0155 |

#### \*\*\*\*\* ANALYSIS NOTES AND ERRORS \*\*\*\*\*

Level of confidence for all confidence intervals in output:

95.0000

Number of bootstrap samples for bias-corrected bootstrap confidence intervals:

5000

----- END MATRIX -----